Credit Card Fraud Detection System using Machine Learning and Generative AI

Project Overview

This project detects fraudulent credit card transactions using a Random Forest classifier trained on the popular credit card transactions dataset. The workflow includes data loading, preprocessing, exploratory analysis, model training, evaluation, confusion matrix visualization, and feature importance analysis.

Role of Generative AI

Generative AI can enhance a fraud detection system by automatically generating synthetic fraud examples to improve model training, explaining why a transaction was classified as fraudulent in natural language, generating investigation summaries for analysts, assisting with anomaly exploration, creating documentation, and supporting conversational fraud analysis through AI assistants.

Technologies Used

Programming: Python
Libraries: Pandas, Matplotlib, Scikit-learn
Model: RandomForestClassifier
Evaluation: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix
Generative AI: Natural language explanations, synthetic data generation, automated reporting, analyst assistance.

Workflow

1. Load dataset
2. Explore and visualize data
3. Split training/testing data
4. Train Random Forest model
5. Predict fraud transactions
6. Evaluate performance
7. Visualize confusion matrix and feature importance
8. Use Generative AI to explain predictions and generate reports.

Python Code (excerpt)

```
import pandas as pd
import matplotlib.pyplot as plt
# ... (same code provided by the user)
# RandomForest-based Credit Card Fraud Detection
```